Practice workbook · Technology, the environment and health

Energy use, resource extraction, e-waste and actions to reduce them, plus health effects/benefits of technology such as wearables.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T56-14

1. Question 5 · part c · 6 marks

2025 JUN · Paper 01

Worked-example retry · Attempt part c only.

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5 Smartwatches are a form of wearable technology.

(a) Embedded systems are used in smartwatches.

(i) A smartwatch has a step tracker that uses a built-in accelerometer sensor.

Complete the table by adding **one other** built-in sensor a smartwatch could use and its purpose.

(2)

Sensor	Purpose
Accelerometer	To measure body movement and step count

(ii) One characteristic of an embedded system in a smartwatch is that it uses sensors.

Give **two other** characteristics of an embedded system that make it suitable for use in a smartwatch.

(2)

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1 .....

.....

2 .....

.....

(b) A software development team writes the software for a smartwatch using assembly language.

(i) Identify the purpose of an assembler.

(1)

- ☒ **A** To convert a low-level language into a high-level language
- ☒ **B** To convert low-level source code into machine code
- ☒ **C** To execute machine code
- ☒ **D** To translate a high-level language into a low-level language

1. Question 5 · part c · 6 marks

2025 JUN · Paper 01

Worked-example retry · Attempt part c only.

(ii) Explain **one** benefit of writing the software for the smartwatch in a low-level language.

(2)

(iii) Members of the software development team are computing professionals.

Give **two** ways that members of the team can demonstrate ethical behaviour.

(2)

1 .....

2 .....

(iv) The software for the smartwatch takes up 500,000 bytes of storage.

Identify the value that is equivalent to 500,000 bytes.

(1)

- ☒ **A** 0.5 MB
- ☒ **B** 500 KiB
- ☒ **C** 0.5 MiB
- ☒ **D** 5 MB

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P 7 8 6 3 7 A 0 1 4 2 0

1. Question 5 · part c · 6 marks

2025 JUN · Paper 01

Worked-example retry · Attempt part c only.

(c) Wearable technology, such as smartwatches, has become very popular with many groups of people including athletes.

Discuss how the use of wearable technology can be a benefit to health.

(6)

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(Total for Question 5 = 16 marks)

1. Question 5 · part c · 6 marks · Working and answer

Working and answer

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2. Question 6 · part c · 6 marks

2021 JUN · Paper 01

Further practice · Attempt part c only.

6 (a) Two emerging technologies are DNA computing and quantum computing.

(i) Explain **one** difference between DNA computing and traditional computing.

(2)

(ii) Give a definition of the term **superposition** in quantum computing.

(1)

(b) Ethically, everyone should be able to access the benefits of technology.

Give **two** reasons why this may not be possible.

(2)

1

2

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16

P 6 6 4 8 9 A 0 1 6 2 0

2. Question 6 · part c · 6 marks

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Further practice · Attempt part c only.

(c) Widespread use of computer technology has a negative environmental impact.

Discuss the negative environmental impact of using computers and actions that could be taken to reduce it.

(6)

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(Total for Question 6 = 11 marks)

2. Question 6 · part c · 6 marks · Working and answer

Working and answer

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3. Question 3 · part e.i · 1 mark

2021 NOV · Paper 01

Further practice · Attempt part e.i only.

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3 Carlo uses his smartphone to make hands-free calls from the car.

(a) The smartphone connects to the car wirelessly forming a network.

Complete the table by adding a tick (✓) to indicate the network type and a tick (✓) to indicate the usage model the network uses.

(2)

Network type	Tick (✓)	Usage model	Tick (✓)
Local Area Network (LAN)		Client-Server	
Wide Area Network (WAN)		Peer-to-peer	
Personal Area Network (PAN)			

(b) Data on a network is transmitted between devices using packets and the TCP/IP layered protocol stack.

Complete the table by naming **two** layers of the stack and giving **one** function for each of the layers that you have named.

(4)

	Layer name	Function
1		
2		

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3. Question 3 · part e.i · 1 mark

2021 NOV · Paper 01

Further practice · Attempt part e.i only.

(c) The smartphone can communicate using Wi-Fi.

Identify which **one** of these is a radio frequency used by smartphones to connect to Wi-Fi.

(1)

- ☒ **A** 2.4 GHz
- ☒ **B** 3 KHz
- ☒ **C** 4.1 GHz
- ☒ **D** 5 KHz

(d) Carlo's smartphone can use 3G, 4G and 5G mobile communication standards.

(i) Give **one** advantage of using a higher frequency band for mobile communications.

(1)

(ii) Explain **one** benefit to Carlo of using 5G rather than 3G.

(2)

(e) Carlo wants to reduce the environmental impact of his smartphone.

(i) Give **one** way Carlo could reduce energy consumption.

(1)

(ii) Explain **one** way, other than reducing energy consumption, that Carlo could reduce the environmental impact of his smartphone.

(2)

(Total for Question 3 = 13 marks)

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3. Question 3 · part e.i · 1 mark · Working and answer

Working and answer

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4. Question 3 · part e.ii · 2 marks

2021 NOV · Paper 01

Further practice · Attempt part e.ii only.

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1		
2		

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4. Question 3 · part e.ii · 2 marks

2021 NOV · Paper 01

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(Total for Question 3 = 13 marks)

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4. Question 3 · part e.ii · 2 marks · Working and answer

Working and answer

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